

SAPIENZA
UNIVERSITÀ DI ROMA

Combinatorial Optimization

Exercises 6 — Specialized algorithms

Worked exercises on the chapters of Volume 6 (specialized algorithms), each followed by its complete solution. The instances are those of the problem catalogue, cited by their code.

Fabio Furini

fabiofurini.github.io

Department of Computer, Control and Management Engineering — Sapienza University of Rome

Online version and downloadable material:
fabiofurini.github.io/combinatorial-optimization/
github.com/fabiofurini/combinatorial-optimization

Academic year 2026–2027

Contents

29 Knapsack: branch-and-bound on the critical item **2**

Chapter 29

Knapsack: branch-and-bound on the critical item

Exercise 29.1 — Branch-and-Bound Algorithm for the Knapsack Problem ($P1$)

We consider a knapsack capacity $c = 11$ and a set $\mathscr{J} = \{1, 2, 3, 4, 5\}$ of 5 items ($n = 5$) with the following profits and weights:

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
p_j	8	6	14	6	2
w_j	3	3	9	5	2

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;
   (d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

   $$\bar{x}_1 = 1, \quad \bar{x}_2 = 1, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1, \quad \bar{x}_5 = 0$$

   and set the Lower Bound LB accordingly.
3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{8}{3} \geq \frac{6}{3} \geq \frac{14}{9} \geq \frac{6}{5} \geq \frac{2}{2}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{lrlrlrlrlrcl} \max & 8 & x_1 & +6 & x_2 & +14 & x_3 & +6 & x_4 & +2 & x_5 & \\ & 3 & x_1 & +3 & x_2 & +9 & x_3 & +5 & x_4 & +2 & x_5 & \leq & 11 \\ & & x_1, & & x_2, & & x_3, & & x_4, & & x_5 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{lrlrlrlrlrcl} \max & 8 & x_1 & +6 & x_2 & +14 & x_3 & +6 & x_4 & +2 & x_5 & & \\ & 3 & x_1 & +3 & x_2 & +9 & x_3 & +5 & x_4 & +2 & x_5 & \leq & 11 \\ & & x_1 & & & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & & & \leq & 1 \\ & & & & & & x_3 & & & & & \leq & 1 \\ & & & & & & & & x_4 & & & \leq & 1 \\ & & & & & & & & & & x_5 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 3$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{5}{9}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 8 \cdot 1 + 6 \cdot 1 + 14 \cdot \frac{5}{9} = \frac{196}{9}$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 20 ($LB(\bar{\boldsymbol{x}}) = 20$).

  Due to the current Lower Bound LB and Upper Bound UB, we have:

$$20 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{196}{9} \right\rfloor}_{=21}$$

- **Iteration 1**

  The LP^1_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_3. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

  (a) Solution of the LP^2_{KP} problem

  We have

  $$\mathscr{I}^2 = \{1, 2, 4, 5\}, \quad \mathscr{I}^2_{\text{out}} = \{3\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

  The capacity is $c(2) = 11$ and the objective-function constant is $p(2) = 0$. The split item is $s = 5$ and the residual capacity is $\bar{c} = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

  $$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = 1, \quad \tilde{x}_5 = 0, \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 20$$

  The LP^2_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^2_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 20 \leq 20$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^3_{KP} problem

  We have

  $$\mathscr{I}^3 = \{1, 2, 4, 5\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{3\}$$

  The capacity is $c(3) = 2$ and the objective-function constant is $p(3) = 14$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 2$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

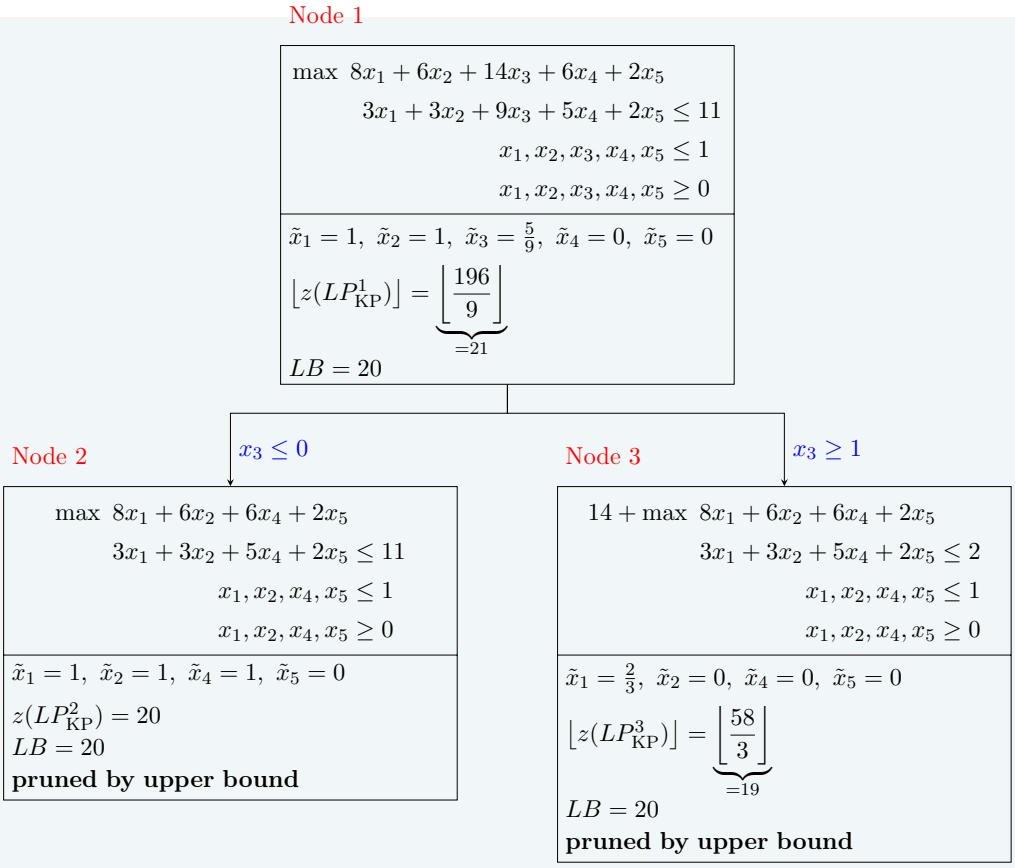
  $$\tilde{x}_1 = \frac{2}{3}, \quad \tilde{x}_2 = 0, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 14 + 8 \cdot \frac{2}{3} = \frac{58}{3}$$

  The LP^3_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^3_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{58}{3} \right\rfloor}_{=19} \leq 20$$

  accordingly, the branching node is pruned by upper bound.

  At the end of the first iteration, Node 2 and Node 3 are both pruned by upper bound. The corresponding Branch-and-Bound Tree is:

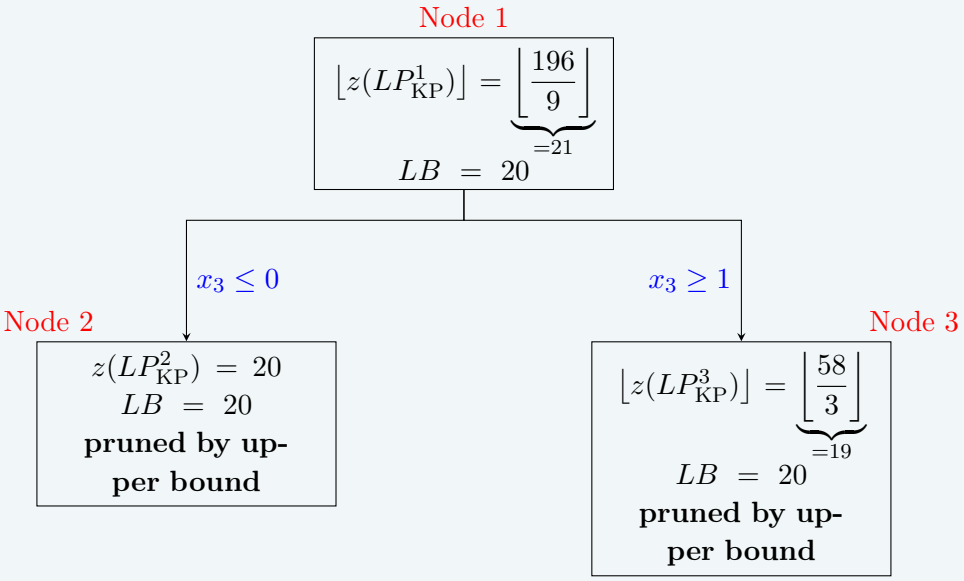


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 20$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 1, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 29.2 — Branch-and-Bound Algorithm for the Knapsack Problem ($P2$)

We consider a knapsack capacity $c = 8$ and a set $\mathscr{J} = \{1, 2, 3, 4\}$ of 4 items ($n = 4$) with the following profits and weights:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
p_j	40	42	25	12
w_j	4	7	5	3

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;
   (d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :
   $$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1$$
   and set the Lower Bound LB accordingly.
3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{40}{4} \geq \frac{42}{7} \geq \frac{25}{5} \geq \frac{12}{3}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 40 & x_1 & +\,42 & x_2 & +\,25 & x_3 & +\,12 & x_4 & & \\ & 4 & x_1 & +\,7 & x_2 & +\,5 & x_3 & +\,3 & x_4 & \leq & 8 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 40 & x_1 & +\,42 & x_2 & +\,25 & x_3 & +\,12 & x_4 & & \\ & 4 & x_1 & +\,7 & x_2 & +\,5 & x_3 & +\,3 & x_4 & \leq & 8 \\ & & x_1 & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & \leq & 1 \\ & & & & & & x_3 & & & \leq & 1 \\ & & & & & & & & x_4 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 2$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{4}{7}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 40 \cdot 1 + 42 \cdot \frac{4}{7} = 64$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 52 ($LB(\bar{\boldsymbol{x}}) = 52$).

  Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \leq z(\, ILP\,) \leq 64$$

- **Iteration 1**

  The LP^1_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_2. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_2 \leq 0$) and selecting ($x_2 \geq 1$) item 2.

  (a) Solution of the LP^2_{KP} problem

  We have

  $$\mathscr{I}^2 = \{1, 3, 4\}, \quad \mathscr{I}^2_{\text{out}} = \{2\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

  The capacity is $c(2) = 8$ and the objective-function constant is $p(2) = 0$. The split item is $s = 3$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

  $$\tilde{x}_1 = 1, \quad \tilde{x}_3 = \frac{4}{5}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 40 \cdot 1 + 25 \cdot \frac{4}{5} = 60$$

  The LP^2_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^2_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad 60 > 52$$

  The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

  (b) Solution of the LP^3_{KP} problem

  We have

  $$\mathscr{I}^3 = \{1, 3, 4\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{2\}$$

  The capacity is $c(3) = 1$ and the objective-function constant is $p(3) = 42$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

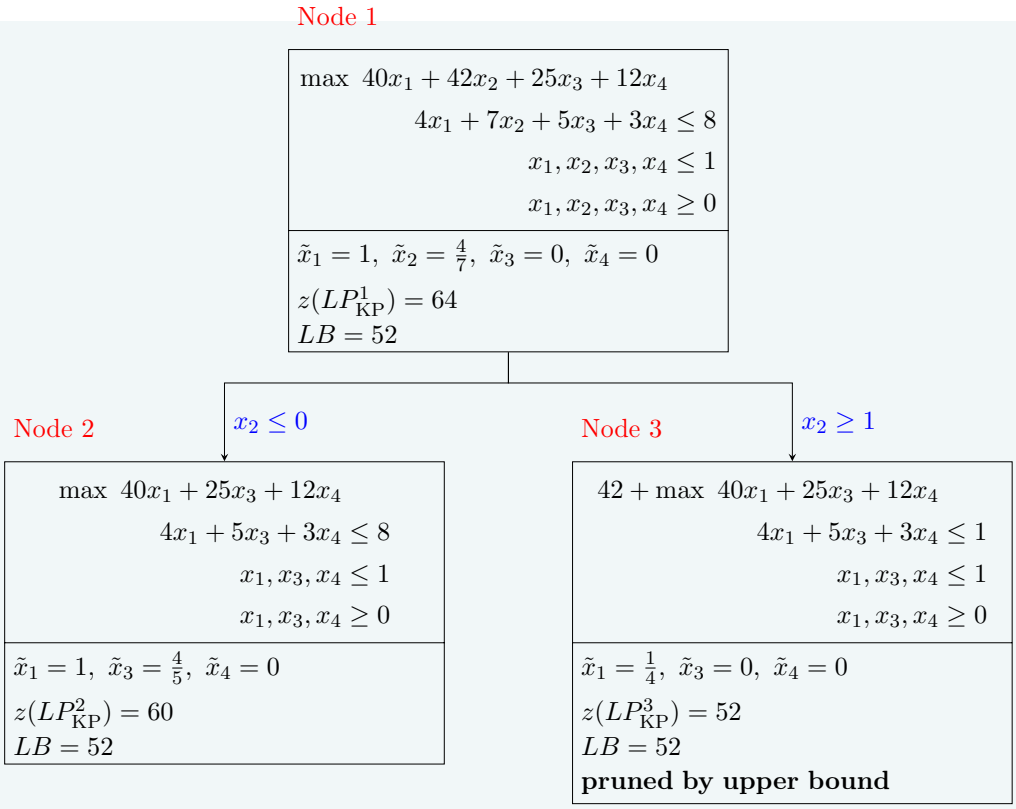
  $$\tilde{x}_1 = \frac{1}{4}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 42 + 40 \cdot \frac{1}{4} = 52$$

  The LP^3_{KP} problem is **feasible** and we have:

  $$\lfloor z(LP^3_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 52 \leq 52$$

  accordingly, the branching node is pruned by upper bound.

  At the end of the first iteration, Node 2 is inserted in the Problem List L, while Node 3 is pruned by upper bound. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \quad \leq \quad z(\ ILP\ ) \quad \leq \quad 60$$

- **Iteration 2**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{1, 4\}, \quad \mathscr{I}^4_{\text{out}} = \{2, 3\}, \quad \mathscr{I}^4_{\text{in}} = \emptyset$$

The capacity is $c(4) = 8$ and the objective-function constant is $p(4) = 0$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\ LP^4_{\text{KP}}\ )$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_4 = 1 \quad \text{and} \quad z(\ LP^4_{\text{KP}}\ ) = 40 \cdot 1 + 12 \cdot 1 = 52$$

The LP^4_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^4_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 52 \leq 52$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^5_{KP} problem

We have

$$\mathscr{I}^5 = \{1, 4\}, \quad \mathscr{I}^5_{\text{out}} = \{2\}, \quad \mathscr{I}^5_{\text{in}} = \{3\}$$

The capacity is $c(5) = 3$ and the objective-function constant is $p(5) = 25$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 3$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^5_{\text{KP}}\,)$ is:

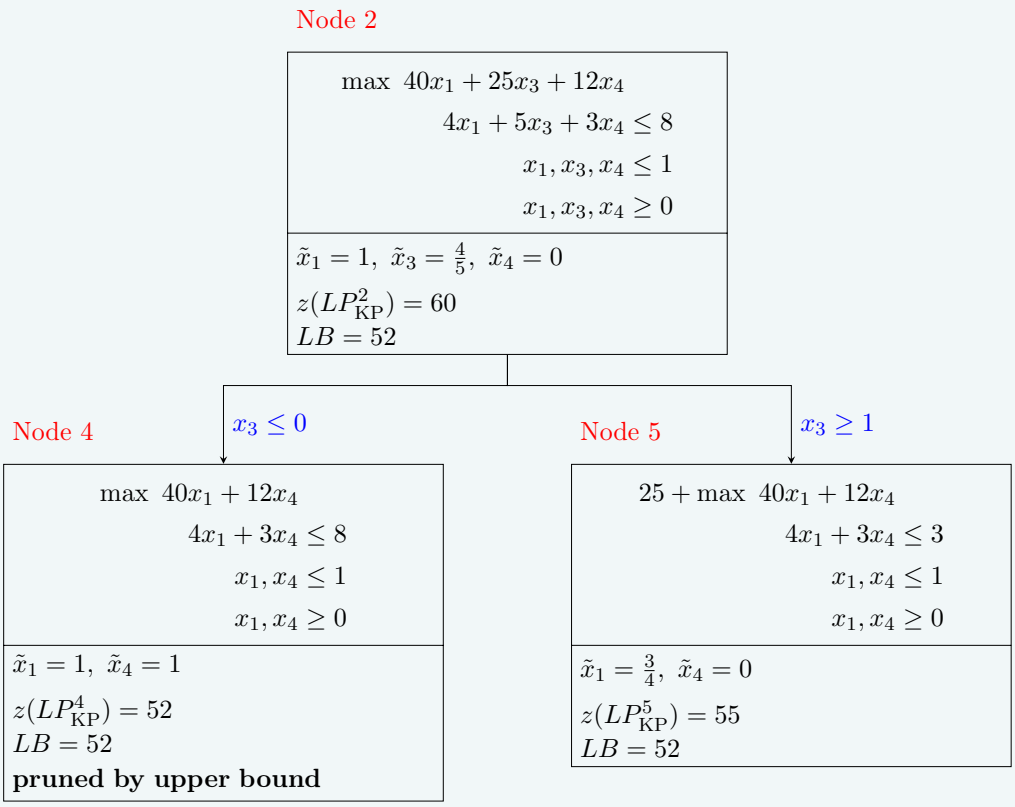
$$\tilde{x}_1 = \frac{3}{4}, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 1, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^5_{\text{KP}}\,) = 25 + 40 \cdot \frac{3}{4} = 55$$

The LP^5_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^5_{\text{KP}}) \right\rfloor > LB \qquad \text{since} \qquad 55 > 52$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^5_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^5_{KP} problem is **inserted in the problem list** L.

At the end of the second iteration, Node 5 is inserted in the Problem List L, while Node 4 is pruned by upper bound. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$52 \quad \leq \quad z(\, ILP\,) \quad \leq \quad 55$$

- **Iteration 3**

  The LP^5_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_1. The LP^6_{KP} and LP^7_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

  (a) Solution of the LP^6_{KP} problem

  We have

  $$\mathscr{I}^6 = \{4\}, \quad \mathscr{I}^6_{\text{out}} = \{1, 2\}, \quad \mathscr{I}^6_{\text{in}} = \{3\}$$

  The capacity is $c(6) = 3$ and the objective-function constant is $p(6) = 25$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^6_{\text{KP}}\,)$ is:

  $$\tilde{x}_4 = 1 \quad \text{and} \quad z(\, LP^6_{\text{KP}}\,) = 25 + 12 \cdot 1 = 37$$

  The LP^6_{KP} problem is **feasible** and we have:

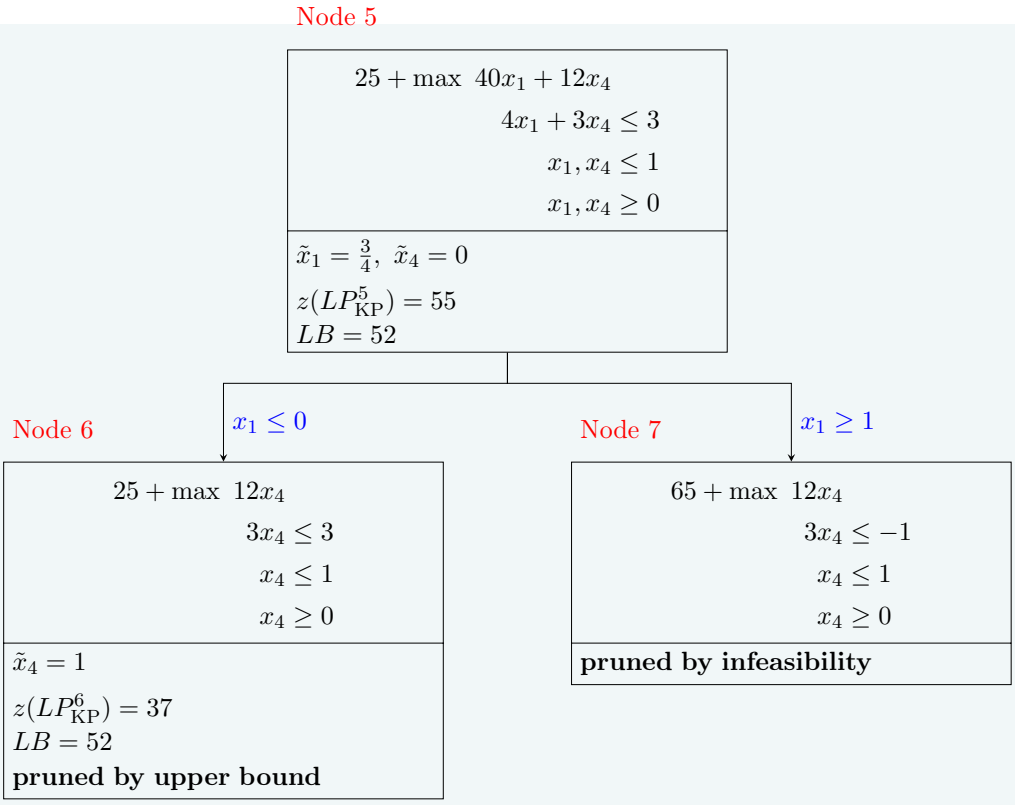
  $$\left\lfloor z(LP^6_{\text{KP}}) \right\rfloor \leq LB \qquad \text{since} \qquad 37 \leq 52$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^7_{KP} problem

  The LP^7_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

  At the end of the third iteration, Node 6 is pruned by upper bound and Node 7 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:

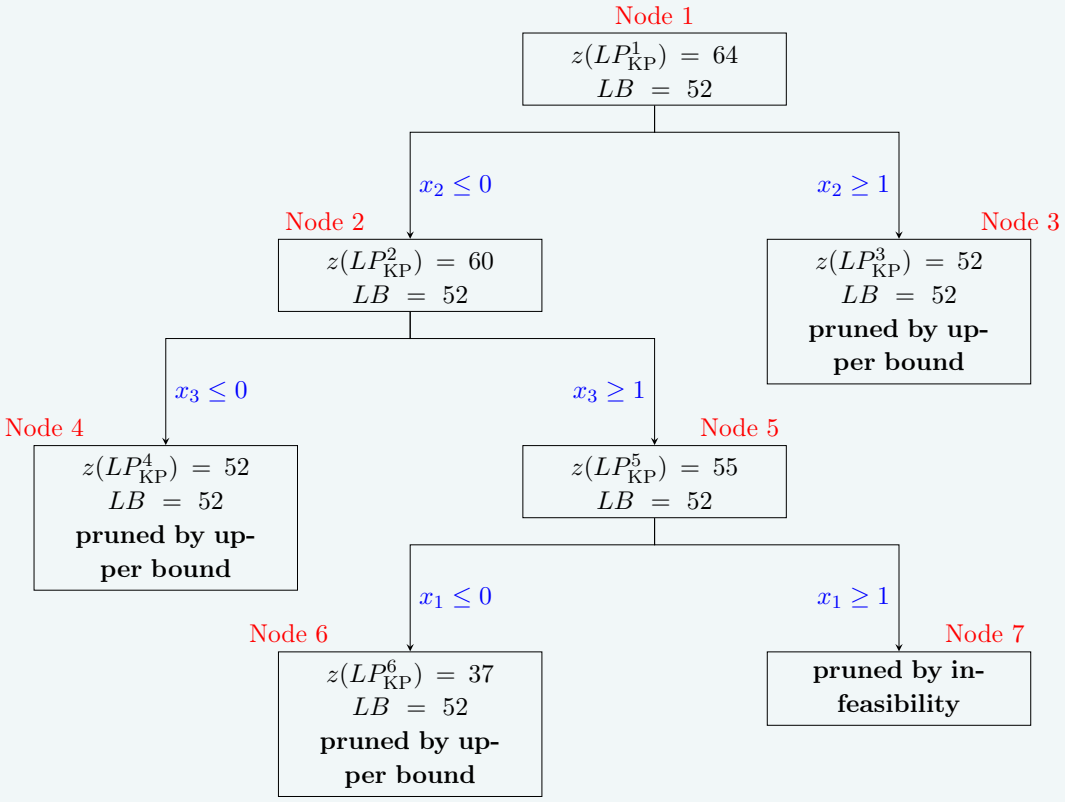


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 52$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 29.3 — Branch-and-Bound Algorithm for the Knapsack Problem ($P3$)

We consider a knapsack capacity $c = 15$ and a set $\mathscr{I} = \{1, 2, 3, 4\}$ of 4 items ($n = 4$) with the following profits and weights:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
p_j	24	17	12	6
w_j	10	8	6	5

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the LIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;

(d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

$$\bar{x}_1 = 1, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 1$$

and set the Lower Bound LB accordingly.

3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{24}{10} \geq \frac{17}{8} \geq \frac{12}{6} \geq \frac{6}{5}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 24 & x_1 & +\,17 & x_2 & +\,12 & x_3 & +\,6 & x_4 & & \\ & 10 & x_1 & +\,8 & x_2 & +\,6 & x_3 & +\,5 & x_4 & \leq & 15 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \in & \{0,1\} \end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrr} \max & 24 & x_1 & +\,17 & x_2 & +\,12 & x_3 & +\,6 & x_4 & & \\ & 10 & x_1 & +\,8 & x_2 & +\,6 & x_3 & +\,5 & x_4 & \leq & 15 \\ & & x_1 & & & & & & & \leq & 1 \\ & & & & x_2 & & & & & \leq & 1 \\ & & & & & & x_3 & & & \leq & 1 \\ & & & & & & & & x_4 & \leq & 1 \\ & & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

  For the LP^1_{KP} problem, the split item is $s = 2$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{5}{8}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0$$

  the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 24 \cdot 1 + 17 \cdot \frac{5}{8} = \frac{277}{8}$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 30 ($LB(\bar{\boldsymbol{x}}) = 30$).

Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{277}{8} \right\rfloor}_{=34}$$

- **Iteration 1**

The LP^1_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_2. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_2 \leq 0$) and selecting ($x_2 \geq 1$) item 2.

(a) Solution of the LP^2_{KP} problem

We have

$$\mathscr{I}^2 = \{1, 3, 4\}, \quad \mathscr{I}^2_{\text{out}} = \{2\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

The capacity is $c(2) = 15$ and the objective-function constant is $p(2) = 0$. The split item is $s = 3$ and the residual capacity is $\bar{c} = 5$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_3 = \frac{5}{6}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^2_{\text{KP}}\,) = 24 \cdot 1 + 12 \cdot \frac{5}{6} = 34$$

The LP^2_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^2_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad 34 > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^3_{KP} problem

We have

$$\mathscr{I}^3 = \{1, 3, 4\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{2\}$$

The capacity is $c(3) = 7$ and the objective-function constant is $p(3) = 17$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 7$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

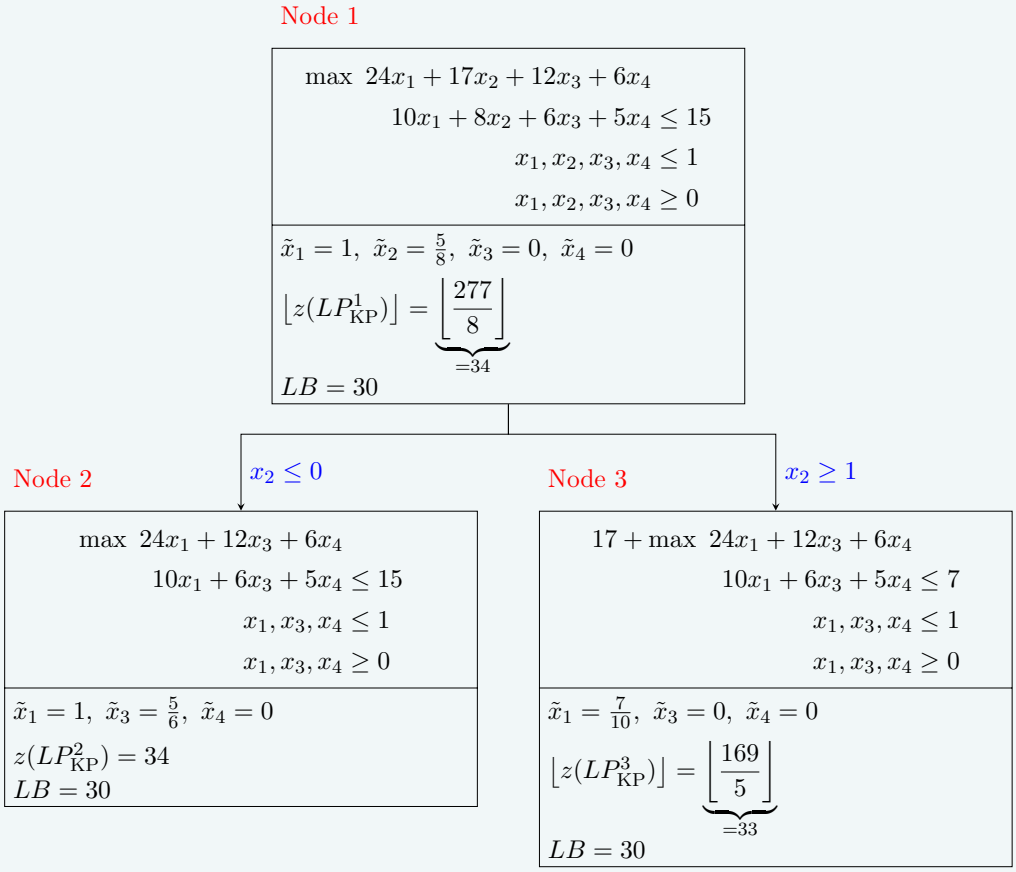
$$\tilde{x}_1 = \frac{7}{10}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^3_{\text{KP}}\,) = 17 + 24 \cdot \frac{7}{10} = \frac{169}{5}$$

The LP^3_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^3_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{169}{5} \right\rfloor}_{=33} > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^3_{KP} problem is **inserted in the problem list** L.

At the end of the first iteration, Node 2 and Node 3 are both inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \;\leq\; z(\; ILP\;) \;\leq\; \underbrace{\max\left\{34,\; \underbrace{\left\lfloor \frac{169}{5} \right\rfloor}_{=33}\right\}}_{=34}$$

- **Iteration 2**

The LP^3_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_1. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{3, 4\}, \quad \mathscr{I}^4_{\text{out}} = \{1\}, \quad \mathscr{I}^4_{\text{in}} = \{2\}$$

The capacity is $c(4) = 7$ and the objective-function constant is $p(4) = 17$. The split item is $s = 4$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^4_{\text{KP}}\,)$ is:

$$\tilde{x}_3 = 1, \quad \tilde{x}_4 = \frac{1}{5} \quad \text{and} \quad z(\, LP^4_{\text{KP}}\,) = 17 + 12 \cdot 1 + 6 \cdot \frac{1}{5} = \frac{151}{5}$$

The LP^4_{KP} problem is **feasible** and we have:

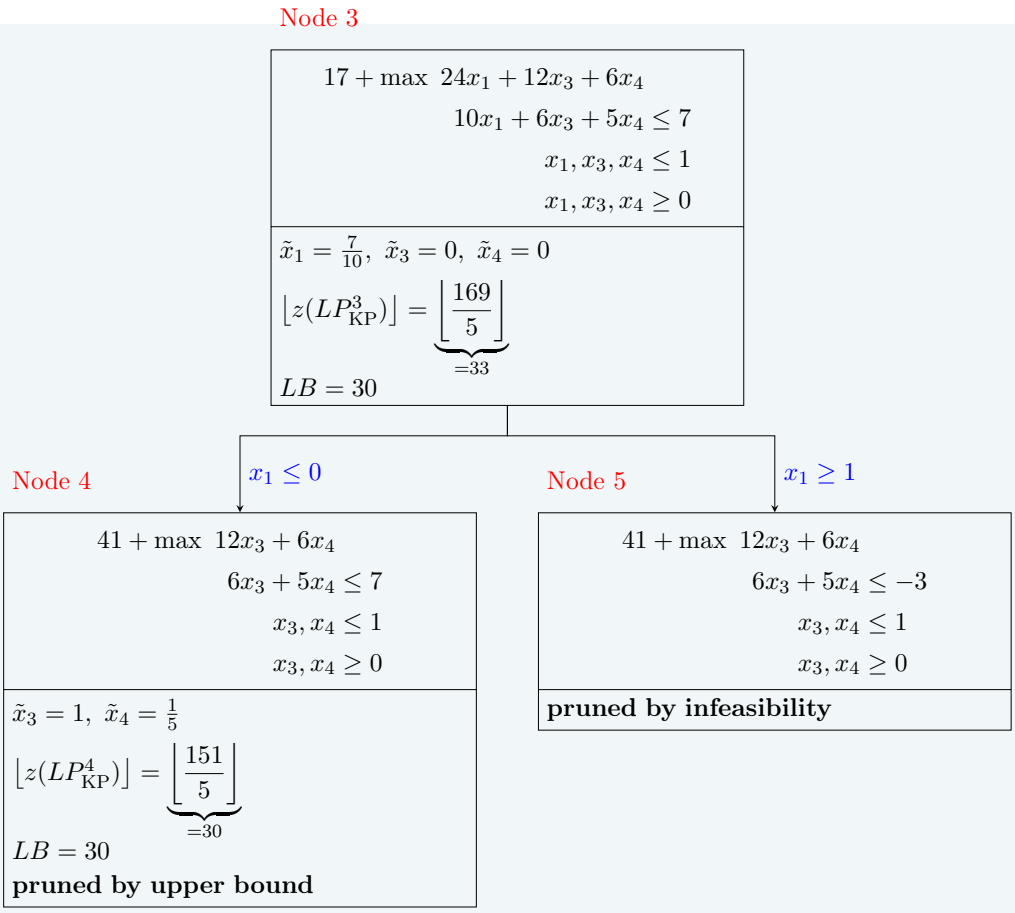
$$\lfloor z(LP^4_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{151}{5} \right\rfloor}_{=30} \leq 30$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^5_{KP} problem

The LP^5_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

At the end of the second iteration, Node 4 is pruned by upper bound and Node 5 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$30 \quad \leq \quad z(\ ILP\ ) \quad \leq \quad 34$$

- **Iteration 3**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^6_{KP} and LP^7_{KP} problems are created by unselecting $(x_3 \leq 0)$ and selecting $(x_3 \geq 1)$ item 3.

(a) Solution of the LP^6_{KP} problem

We have

$$\mathcal{I}^6 = \{1, 4\}, \quad \mathcal{I}^6_{\text{out}} = \{2, 3\}, \quad \mathcal{I}^6_{\text{in}} = \emptyset$$

The capacity is $c(6) = 15$ and the objective-function constant is $p(6) = 0$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\ LP^6_{\text{KP}}\ )$ is:

$$\tilde{x}_1 = 1, \tilde{x}_4 = 1 \quad \text{and} \quad z(\ LP^6_{\text{KP}}\ ) = 24 \cdot 1 + 6 \cdot 1 = 30$$

The LP^6_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^6_{\text{KP}}) \rfloor \leq LB \qquad \text{since} \qquad 30 \leq 30$$

accordingly, the branching node is pruned by upper bound.

(b) Solution of the LP^7_{KP} problem

We have

$$\mathscr{I}^7 = \{1, 4\}, \quad \mathscr{I}^7_{\text{out}} = \{2\}, \quad \mathscr{I}^7_{\text{in}} = \{3\}$$

The capacity is $c(7) = 9$ and the objective-function constant is $p(7) = 12$. The split item is $s = 1$ and the residual capacity is $\bar{c} = 9$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^7_{\text{KP}}\,)$ is:

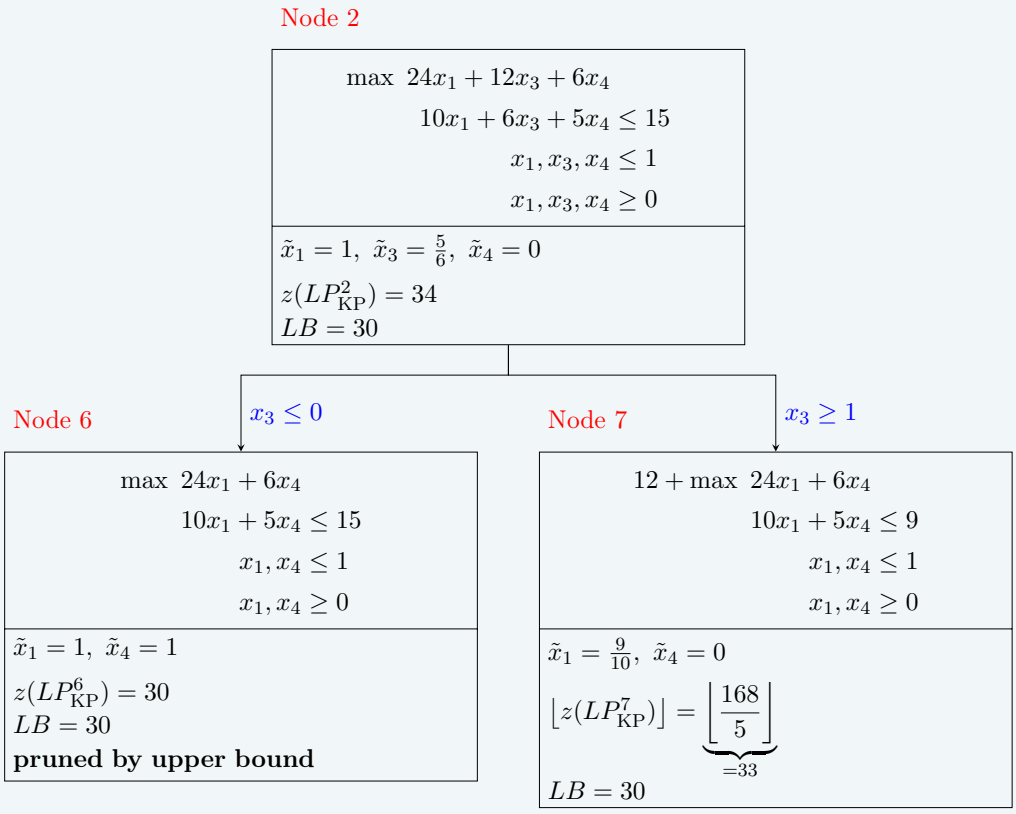
$$\tilde{x}_1 = \frac{9}{10}, \quad \tilde{x}_4 = 0 \quad \text{and} \quad z(\, LP^7_{\text{KP}}\,) = 12 + 24 \cdot \frac{9}{10} = \frac{168}{5}$$

The LP^7_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^7_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{168}{5} \right\rfloor}_{=33} > 30$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^7_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^7_{KP} problem is **inserted in the problem list** L.

At the end of the third iteration, Node 6 is pruned by upper bound and Node 7 is inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



- **Iteration 4**

  The LP^7_{KP} problem is extracted and deleted from the Problem List L.

  The branching variable is x_1. The LP^8_{KP} and LP^9_{KP} problems are created by unselecting ($x_1 \leq 0$) and selecting ($x_1 \geq 1$) item 1.

  (a) Solution of the LP^8_{KP} problem

  We have

  $$\mathscr{I}^8 = \{4\}, \quad \mathscr{I}^8_{\text{out}} = \{1, 2\}, \quad \mathscr{I}^8_{\text{in}} = \{3\}$$

  The capacity is $c(8) = 9$ and the objective-function constant is $p(8) = 12$. All items fit into the knapsack. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^8_{\text{KP}}\,)$ is:

  $$\tilde{x}_4 = 1 \quad \text{and} \quad z(\, LP^8_{\text{KP}}\,) = 12 + 6 \cdot 1 = 18$$

  The LP^8_{KP} problem is **feasible** and we have:

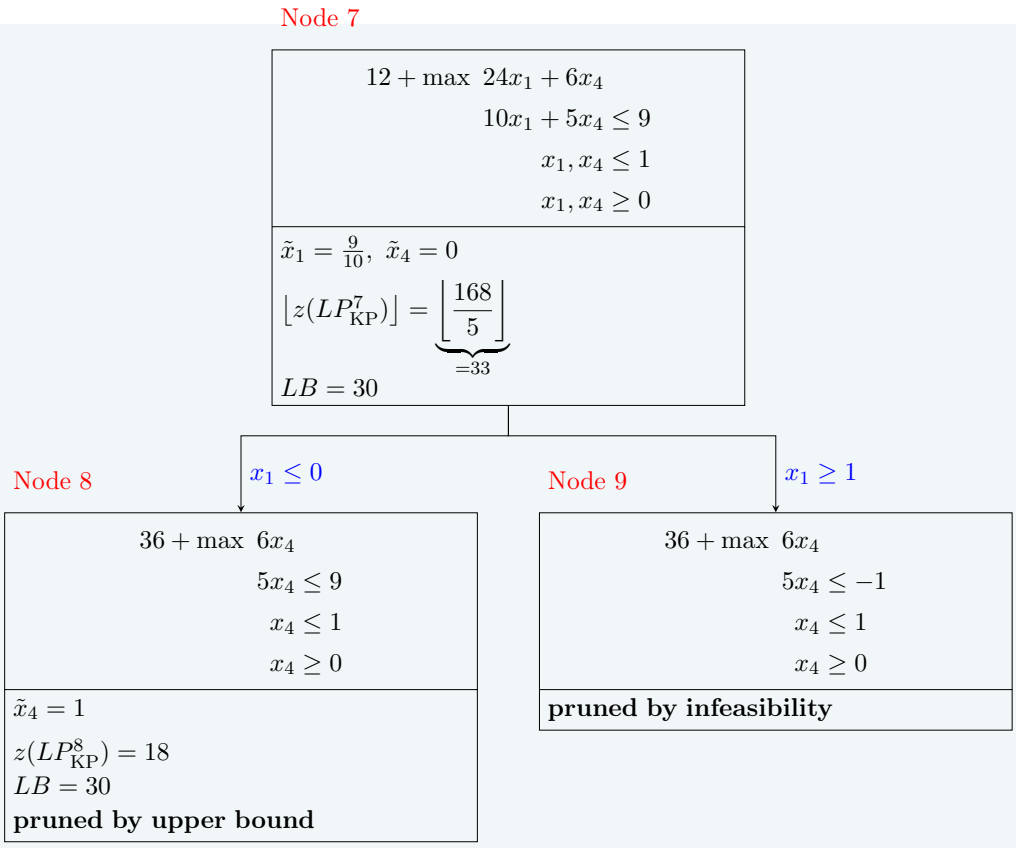
  $$\left\lfloor z(LP^8_{\text{KP}}) \right\rfloor \leq LB \qquad \text{since} \qquad 18 \leq 30$$

  accordingly, the branching node is pruned by upper bound.

  (b) Solution of the LP^9_{KP} problem

  The LP^9_{KP} problem is **infeasible**, accordingly the branching node is pruned by infeasibility.

  At the end of the fourth iteration, Node 8 is pruned by upper bound and Node 9 is pruned by infeasibility. The corresponding Branch-and-Bound Tree is:

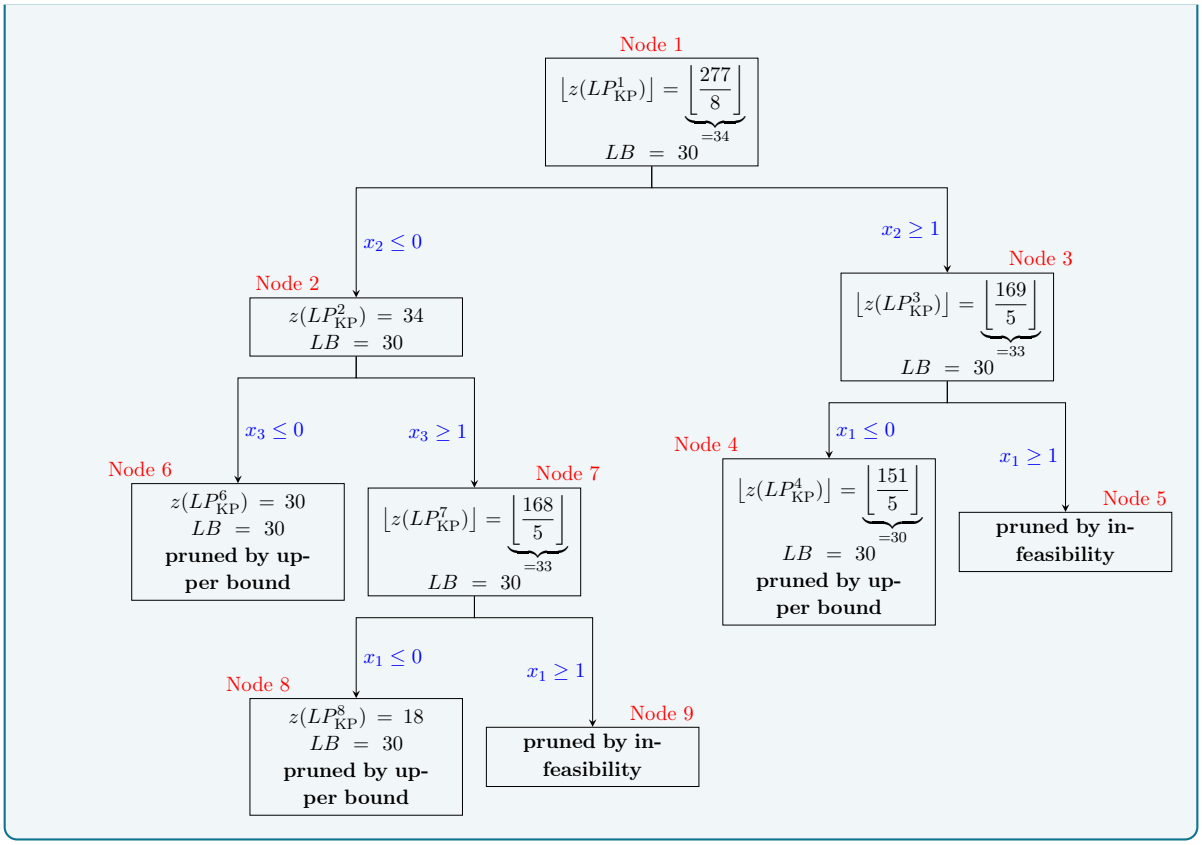


- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 30$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.



Exercise 29.4 — Branch-and-bound Algorithm for the Knapsack Problem ($P4$)

We consider a knapsack capacity $c = 9$ and a set $\mathscr{I} = \{1, 2, 3, 4, 5, 6\}$ of 6 items ($n = 6$) with the following profits and weights:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$
p_i	6	5	8	9	6	7
w_i	2	3	6	7	5	9

Questions

1. Check if the items are sorted by non-increasing efficiency otherwise sort them in this order. Write the ILP_{KP} model and the LP_{KP} model for the KP instance.
2. Execute the branch-and-bound algorithm to solve the ILP_{KP} model for the Knapsack Problem using:
   (a) the split-item rule as branching-variable selection strategy;
   (b) the FIFO rule as branching-node selection strategy;
   (c) solve first the down-branching node and then the up-branching node;

(d) set the Incumbent Solution $\bar{\boldsymbol{x}}(\, ILP_{\text{KP}}\,)$ to the following heuristic solution :

$$\bar{x}_1 = 1, \quad \bar{x}_2 = 1, \quad \bar{x}_3 = 0, \quad \bar{x}_4 = 0, \quad \bar{x}_5 = 0, \quad \bar{x}_6 = 0$$

and set the Lower Bound LB accordingly.

3. Determine and comment the correspondent branch-and-bound tree showing the reason why the branching nodes can be pruned.

Solution

1. The items are sorted by non-increasing efficiency since we have:

$$\frac{6}{2} \geq \frac{5}{3} \geq \frac{8}{6} \geq \frac{9}{7} \geq \frac{6}{5} \geq \frac{7}{9}$$

This is the ILP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrrrrrr}
\max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\
& 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\
& & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \in & \{0,1\}
\end{array}$$

This is the LP_{KP} model for the KP instance:

$$\begin{array}{rrrrrrrrrrrrrrr}
\max & 6 & x_1 & +5 & x_2 & +8 & x_3 & +9 & x_4 & +6 & x_5 & +7 & x_6 & & \\
& 2 & x_1 & +3 & x_2 & +6 & x_3 & +7 & x_4 & +5 & x_5 & +9 & x_6 & \leq & 9 \\
& & x_1 & & & & & & & & & & & \leq & 1 \\
& & & & x_2 & & & & & & & & & \leq & 1 \\
& & & & & & x_3 & & & & & & & \leq & 1 \\
& & & & & & & & x_4 & & & & & \leq & 1 \\
& & & & & & & & & & x_5 & & & \leq & 1 \\
& & & & & & & & & & & & x_6 & \leq & 1 \\
& & x_1, & & x_2, & & x_3, & & x_4, & & x_5, & & x_6 & \geq & 0
\end{array}$$

2. Execution of the branch-and-bound algorithm to solve the ILP_{KP} model.

- **Initialization**

For the LP^1_{KP} problem, the split item is $s = 3$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{2}{3}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0$$

the optimal value is:

$$z(\, LP^1_{\text{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 8 \cdot \frac{2}{3} = \frac{49}{3}$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^1_{\text{KP}}\,)$ is **fractional** and the LP^1_{KP} problem is **feasible**, so the LP^1_{KP} problem is inserted in the Problem List L and the Lower Bound LB is set to 11 ($LB(\bar{\boldsymbol{x}}) = 11$).

Due to the current Lower Bound LB and Upper Bound UB, we have:

$$11 \quad \leq \quad z(\, ILP\,) \quad \leq \quad \underbrace{\left\lfloor \frac{49}{3} \right\rfloor}_{=16}$$

- **Iteration 1**

The LP^1_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_3. The LP^2_{KP} and LP^3_{KP} problems are created by unselecting ($x_3 \leq 0$) and selecting ($x_3 \geq 1$) item 3.

(a) Solution of the LP^2_{KP} problem

We have

$$\mathscr{I}^2 = \{1, 2, 4, 5, 6\}, \quad \mathscr{I}^2_{\text{out}} = \{3\}, \quad \mathscr{I}^2_{\text{in}} = \emptyset$$

The capacity is $c(2) = 9$ and the objective-function constant is $p(2) = 0$. The split item is $s = 4$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_4 = \frac{4}{7}, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \qquad \text{and}$$

$$z(\, LP^2_{\text{KP}}\,) = 6 \cdot 1 + 5 \cdot 1 + 9 \cdot \frac{4}{7} = \frac{113}{7}$$

The LP^2_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^2_{\text{KP}}) \right\rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{113}{7} \right\rfloor}_{=16} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^2_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^2_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^3_{KP} problem

We have

$$\mathscr{I}^3 = \{1,2,4,5,6\}, \quad \mathscr{I}^3_{\text{out}} = \emptyset, \quad \mathscr{I}^3_{\text{in}} = \{3\}$$

The capacity is $c(3) = 3$ and the objective-function constant is $p(3) = 8$. The split item is $s = 2$ and the residual capacity is $\bar{c} = 1$. An optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is:

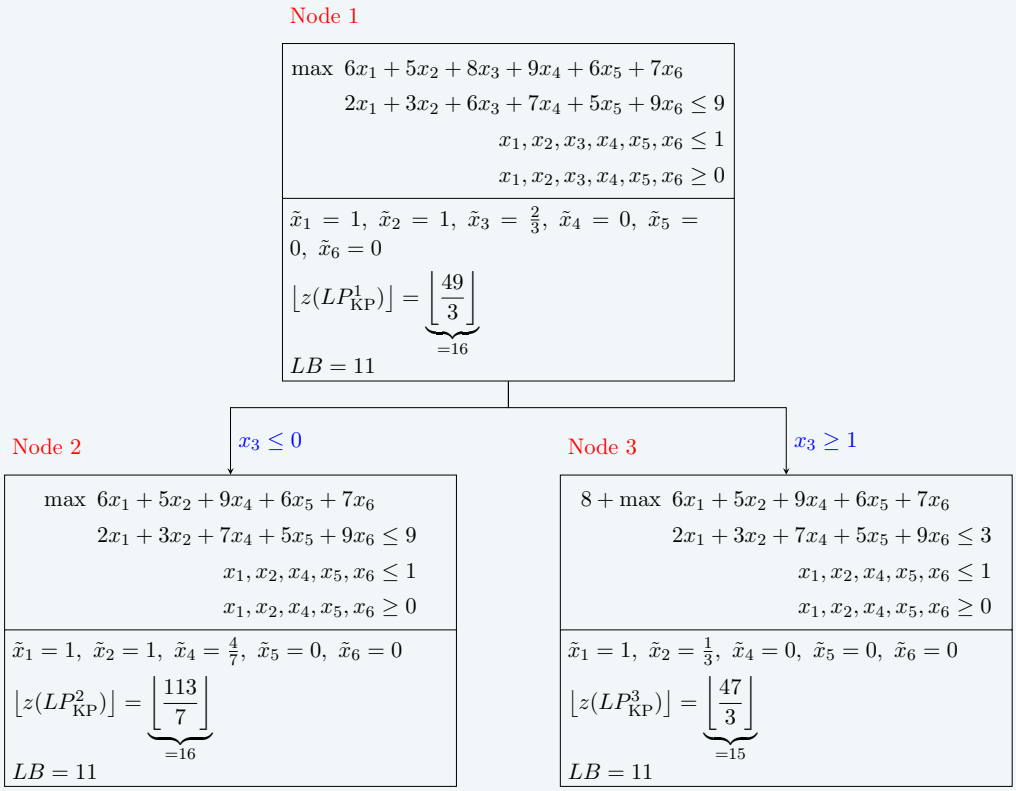
$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{1}{3}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0, \quad \tilde{x}_6 = 0 \qquad \text{and}$$
$$z(\, LP^3_{\text{KP}}\,) = 8 + 6 \cdot 1 + 5 \cdot \frac{1}{3} = \frac{47}{3}$$

The LP^3_{KP} problem is **feasible** and we have:

$$\lfloor z(LP^3_{\text{KP}}) \rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\, LP^3_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^3_{KP} problem is **inserted in the problem list** L.

At the end of the first iteration, Node 2 and Node 3 are both inserted in the Problem List L. The corresponding Branch-and-Bound Tree is:



Due to the current Lower Bound LB and Upper Bound UB, we have:

$$11 \;\leq\; z(\,ILP\,) \;\leq\; \underbrace{\max\left\{\underbrace{\left\lfloor\frac{113}{7}\right\rfloor}_{=16},\; \underbrace{\left\lfloor\frac{47}{3}\right\rfloor}_{=15}\right\}}_{=16}$$

- **Iteration 2**

The LP^2_{KP} problem is extracted and deleted from the Problem List L.

The branching variable is x_4. The LP^4_{KP} and LP^5_{KP} problems are created by unselecting ($x_4 \leq 0$) and selecting ($x_4 \geq 1$) item 4.

(a) Solution of the LP^4_{KP} problem

We have

$$\mathscr{I}^4 = \{1,2,5,6\}, \quad \mathscr{I}^4_{\text{out}} = \{3,4\}, \quad \mathscr{I}^4_{\text{in}} = \emptyset$$

The capacity is $c(4) = 9$ and the objective-function constant is $p(4) = 0$. The split item is $s = 5$ and the residual capacity is $\bar{c} = 4$. An optimal solution $\tilde{\boldsymbol{x}}(\,LP^4_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \;\; \tilde{x}_2 = 1, \;\; \tilde{x}_5 = \frac{4}{5}, \;\; \tilde{x}_6 = 0 \quad \text{and} \quad z(\,LP^4_{\text{KP}}\,) = 6\cdot 1 + 5\cdot 1 + 6\cdot\frac{4}{5} = \frac{79}{5}$$

The LP^4_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^4_{\text{KP}})\right\rfloor > LB \qquad \text{since} \qquad \underbrace{\left\lfloor\frac{79}{5}\right\rfloor}_{=15} > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\,LP^4_{\text{KP}}\,)$ is **fractional**, accordingly, the LP^4_{KP} problem is **inserted in the problem list** L.

(b) Solution of the LP^5_{KP} problem

We have

$$\mathscr{I}^5 = \{1,2,5,6\}, \quad \mathscr{I}^5_{\text{out}} = \{3\}, \quad \mathscr{I}^5_{\text{in}} = \{4\}$$

The capacity is $c(5) = 2$ and the objective-function constant is $p(5) = 9$. The split item is $s = 2$ and the residual capacity is $\bar{c} = 0$. An optimal solution $\tilde{\boldsymbol{x}}(\,LP^5_{\text{KP}}\,)$ is:

$$\tilde{x}_1 = 1, \;\; \tilde{x}_2 = 0, \;\; \tilde{x}_5 = 0, \;\; \tilde{x}_6 = 0 \quad \text{and} \quad z(\,LP^5_{\text{KP}}\,) = 9 + 6\cdot 1 = 15$$

The LP^5_{KP} problem is **feasible** and we have:

$$\left\lfloor z(LP^5_{\text{KP}})\right\rfloor > LB \qquad \text{since} \qquad 15 > 11$$

The optimal solution $\tilde{\boldsymbol{x}}(\,LP^5_{\mathrm{KP}}\,)$ is **integer**, accordingly, the lower bound LB becomes 15 and the incumbent solution $\bar{\boldsymbol{x}}(ILP)$ becomes $\bar{x}_1 = 1, \bar{x}_2 = 0, \bar{x}_3 = 0, \bar{x}_4 = 1, \bar{x}_5 = 0, \bar{x}_6 = 0$.

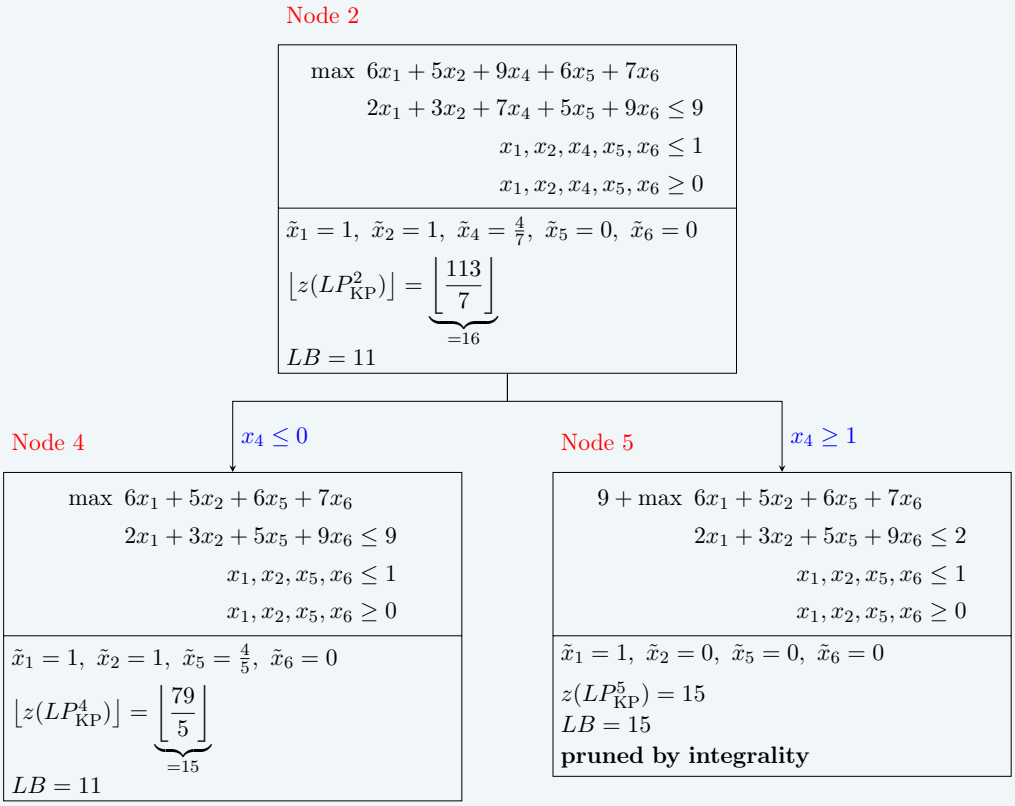
Accordingly, the branching node is pruned by integrality.

Since the Lower Bound is improved, the LP^3_{KP} problem and the LP^4_{KP} problem are pruned by upper bound and removed from the problem list L. We have:

$$\lfloor z(LP^3_{\mathrm{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{47}{3} \right\rfloor}_{=15} \leq 15$$

$$\lfloor z(LP^4_{\mathrm{KP}}) \rfloor \leq LB \qquad \text{since} \qquad \underbrace{\left\lfloor \frac{79}{5} \right\rfloor}_{=15} \leq 15$$

At the end of the second iteration, Node 5 improves the Lower Bound to $LB = 15$ and is pruned by integrality; Node 3 and Node 4 are then pruned by upper bound. The corresponding Branch-and-Bound Tree is:



- **Termination**

  Since the Problem List L is empty, the ILP_{KP} problem is solved to optimality.

  The optimal value is $z(\ ILP_{\text{KP}}\ ) = 15$ (the value of the Lower Bound LB) and an optimal solution $\tilde{\boldsymbol{x}}(\ ILP_{\text{KP}}\ )$ is $\tilde{x}_1 = 1, \tilde{x}_2 = 0, \tilde{x}_3 = 0, \tilde{x}_4 = 1, \tilde{x}_5 = 0, \tilde{x}_6 = 0$ (the Incumbent Solution $\bar{\boldsymbol{x}}(ILP_{\text{KP}})$).

3. This is the **Branch-and-Bound Tree** of the previous execution of the **Branch-and-Bound Algorithm**, showing for each solved branching node the optimal value of the associated LP_{KP} problem, the current Lower Bound LB, and the pruning criteria.

